

بسم الله الرحمن الرحيم

Intestinal protozoa
3- Balantidium Coli

First Course
Lecture 5

Balantidium coli

Objective:

- **Disease:** Balantidiasis or balantidial dysentery (zoonosis disease)
- **Causative agent:** *Balantidium Coli*
- **Infective stage:** mature cyst stage from patient's stool or pigs.
- **Route of entry:** Orally
- **Habitat:** large intestine (colon & caecum).
- **Host:** pigs, rats, monkey and humans.
- **Diagnostic stage:** cyst & trophozoite stage in patient's stool.
- **Natural host:** pig
- Ciliate and largest
- **Accidental host:** human body
- **Reservoir host:** pig, monkey, and rat



Introduction

Balantidium coli belong to the phylum ciliophora and balantididae family.

It is the only ciliate and largest protozoan parasite of humans.

It was first described by Malmsten in 1857 in dysenteric stool of patients

Morphology

B. Coli occurs in two stages (trophozoite and cyst stage), in trophozoite stage it measures about 60-70x40-50 μm . It lives in the large intestine of human, pigs, and monkeys feeding on cell debris, bacteria, starch grains, and other particles. The trophozoite is actively motile by cilia which is short delicate cilia of various lengths over the interior surface of the body. Its anterior end is narrow and posterior end is broad.

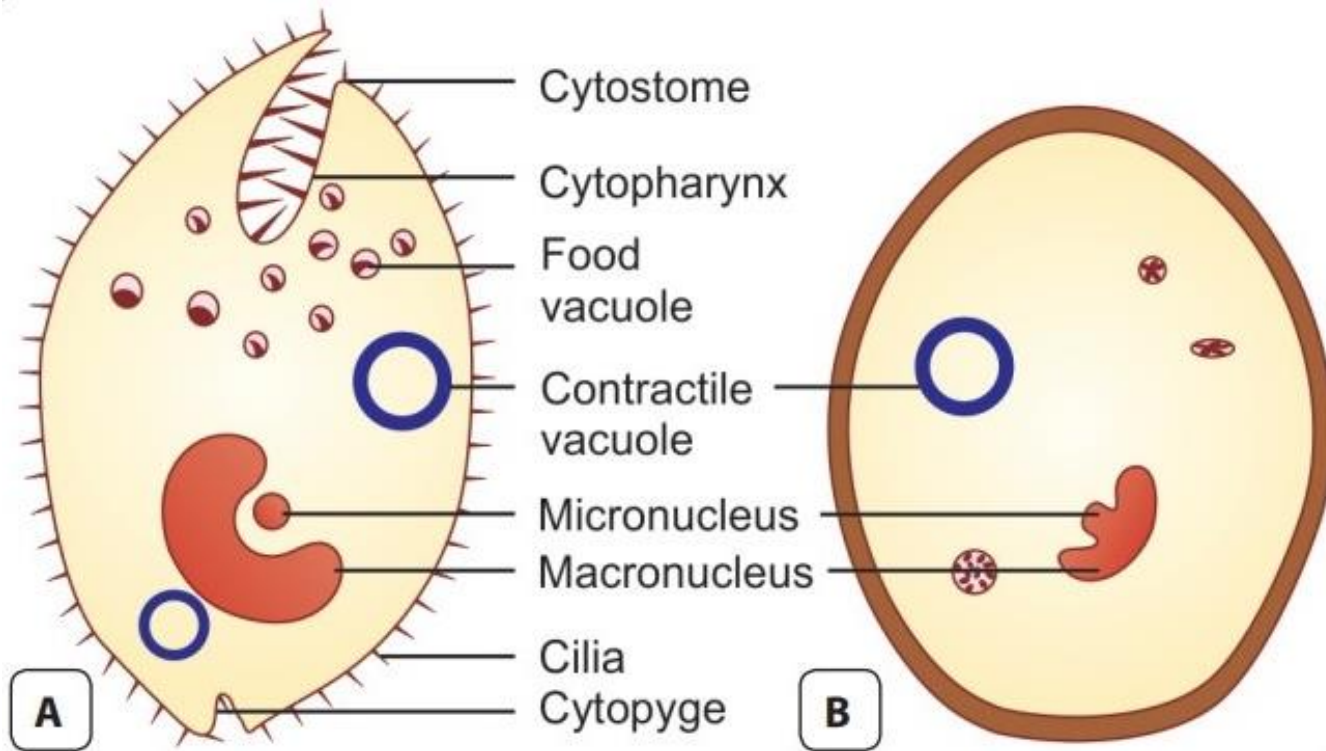
At The interior end, there is a groove leading to the mouth called cytostome and short funnel shaped gullet called cytopharynx. Posteriorly there is a small anal pore called cytopyge. Trophozoite have two nucleus, large kidney shaped called macronucleus and small spherical shaped called micronucleus, cytoplasm has one or two contractile vacuoles and several food vacuoles, with granules of starch, cellular debris, bacteria, erythrocytes etc.



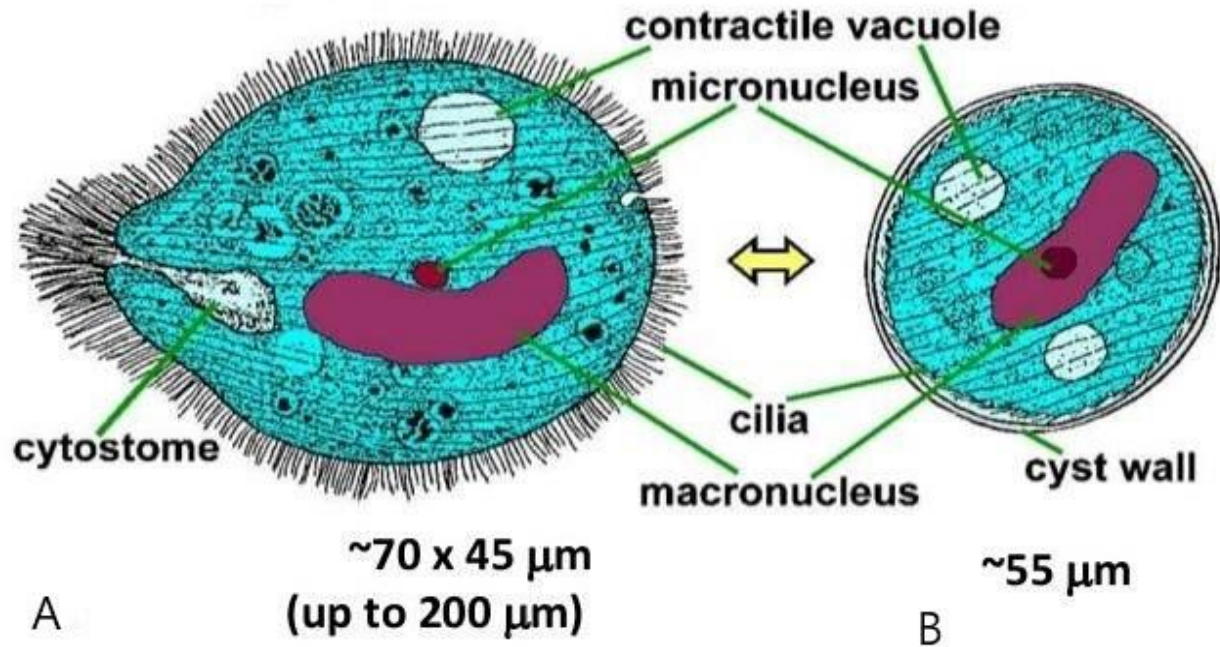
Cyst stage is the infective stage, spherical in shape and measures 40-60 μm in diameter.

It is surrounded by a thick and transparent double layered wall that protect them from external unsuitable conditions with endo cilia.

Cytoplasm is granular with macro and micronucleus. It is found mainly in chronic diseases and carriers, and also found in stool of pigs, monkeys, or human, and is transmitted to the human from digestive tract by reproduction.



Morphology of *B. coli*, A. Trophozoite stage, B. Cyst stage



Morphology of *B. coli*, A. Trophozoite stage, B. Cyst stage

Life cycle

The life cycle is simple and need one host to complete the life cycle, natural host is pigs and accidental host is humans. Humans acquire the infection when ingestion of food, water, or fingers when contaminated with mature cyst stage (infective stage) from stool of patient's (or carrier & pigs).

When the cyst stage reach stomach the cyst wall resist gastric juice, but if the trophozoite of **B. Coli** are ingested. They will be killed in stomach by action of gastric juice. The cysts pass through the stomach undamaged and enter the cavity of small intestine and start to excystation. There is no nuclear division in cyst stage and remains free in the intestinal cavity, and liberated one small trophozoite from each cyst but don't infect the wall of small intestine and carried to large intestine.



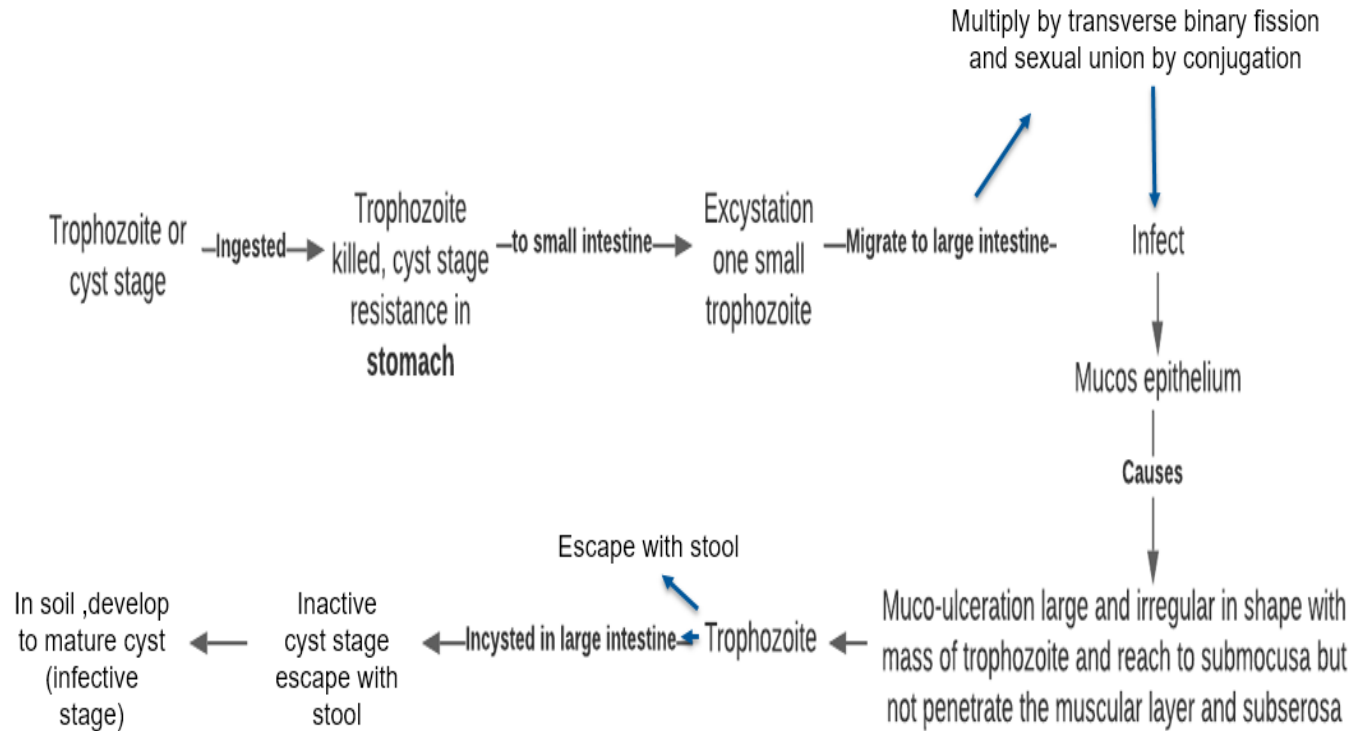
The trophozoites in the cavity of large intestine multiply by transverse binary fission and sexual union by conjugation, take place between two trophozoites and exchange of nuclear material.

Some of trophozoite in cavity of large intestine are passed with stool outside the body, another trophozoite under unsuitable conditions are encysted and become cyst stage (inactive) and pass in the stool outside the body. The cyst stage remain viable in stool for 1-2 days in soil, in suitable conditions, they develop to active (mature cyst).

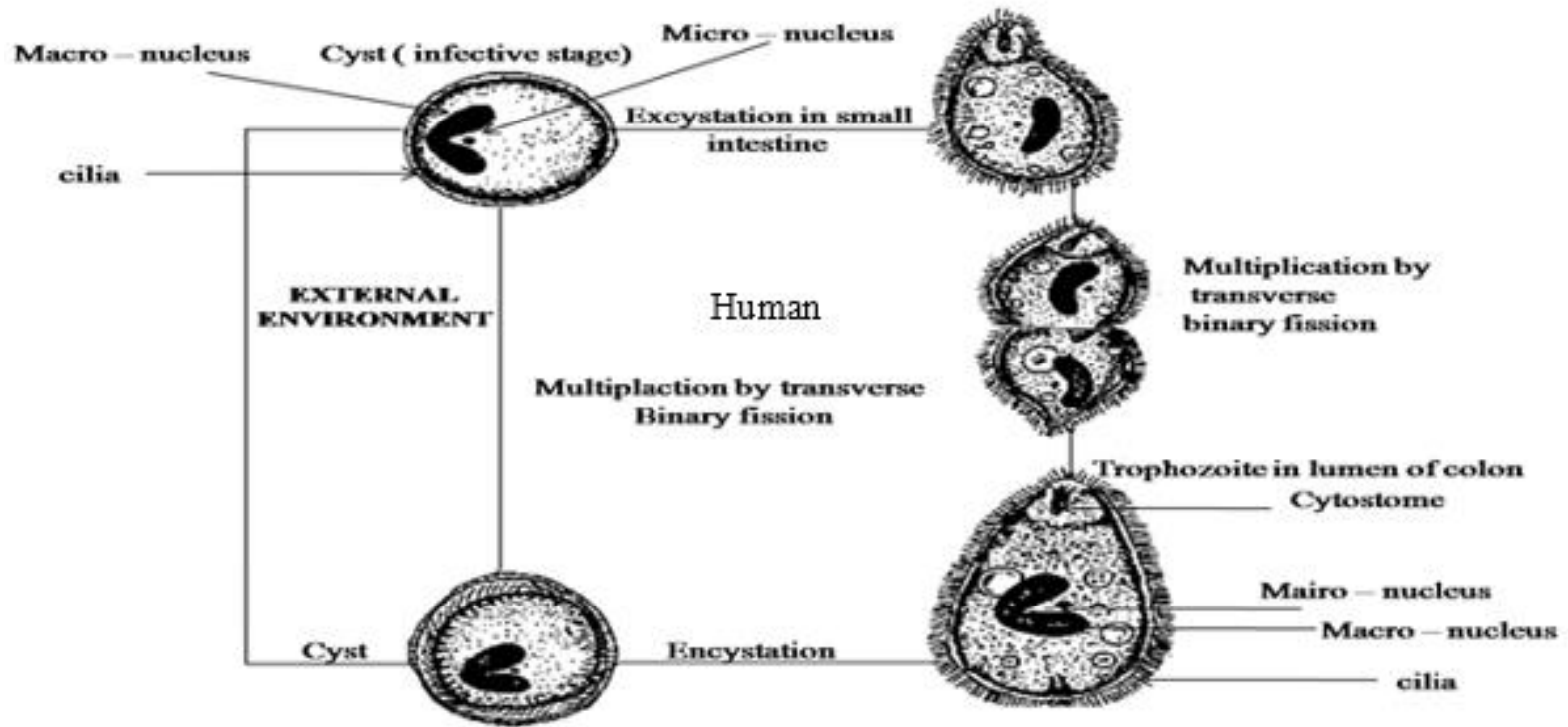


The trophozoite in cavity of large intestine start secreting special enzymes which help in invading the mucous membrane of large intestine, they reach the wall tissue but don't penetrate the wall of large intestine.

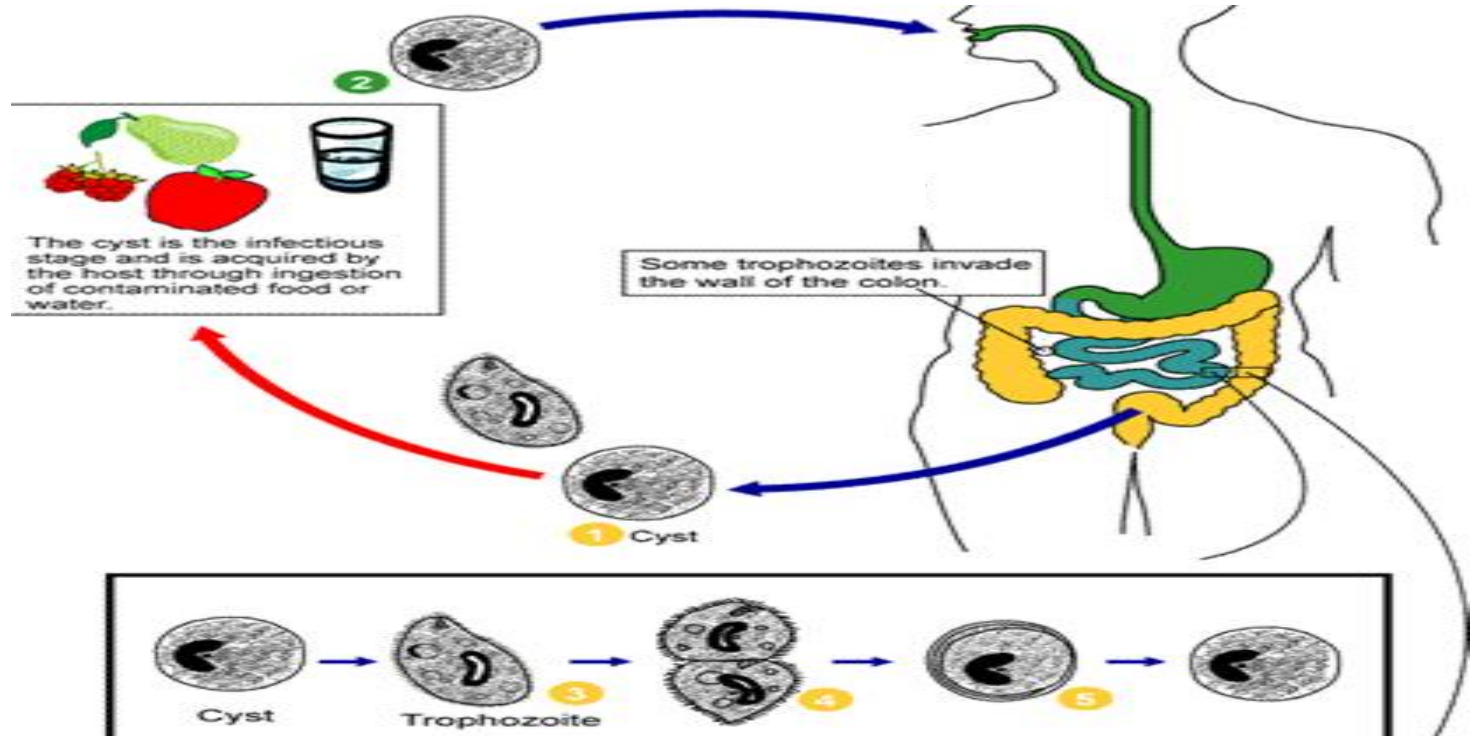
Encystment takes place again as the trophozoites are descending along the colon



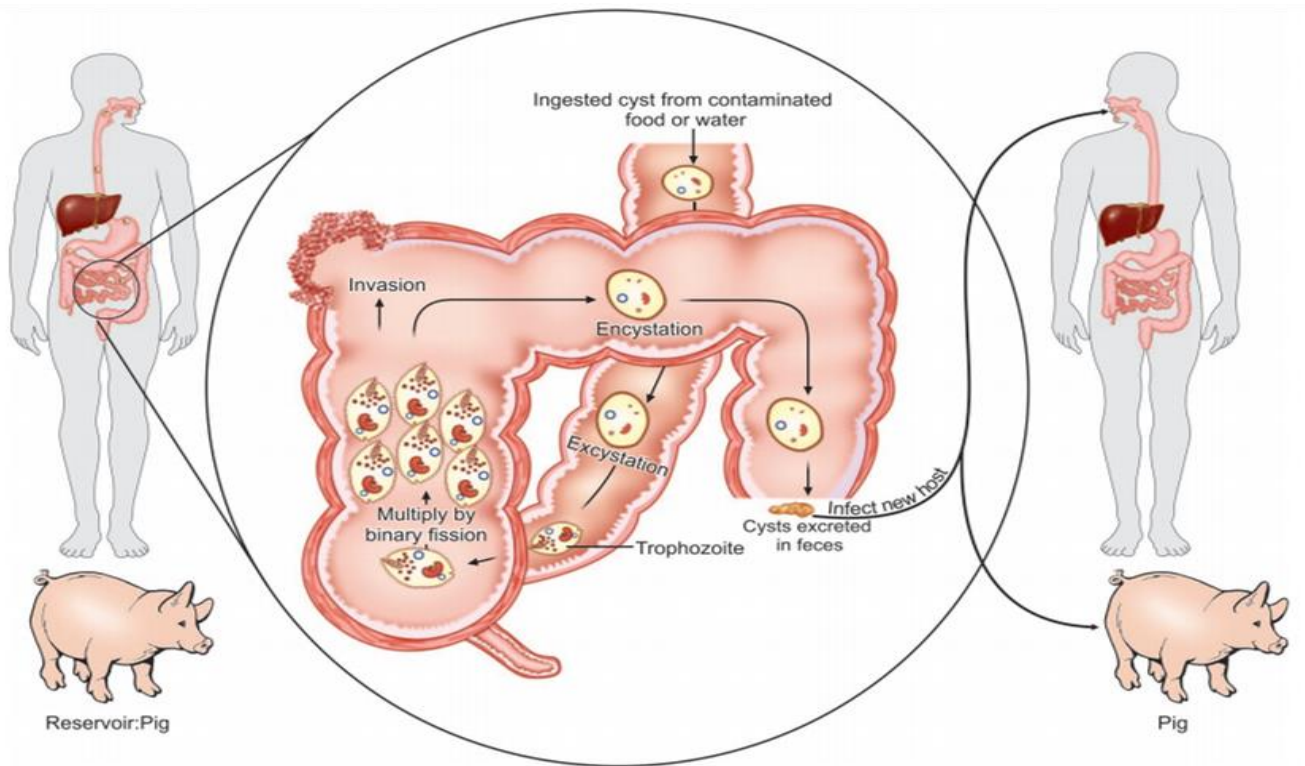
Schematic diagram of life cycle of *B. coli* in human host and soil



Schematic diagram of life cycle of *B. coli* in human host



Life cycle of *B. coli* in human host, multiply by transverse binary fission, and sexual union by conjugation



Life cycle of *B. coli* in human host and reservoir host



Pathogenesis and clinical features

Most cases of **balantidiasis** are asymptomatic. The clinical diagnosis occurs only when the resistance of host is lowered by predisposing factors like malnourishment, alcoholism, achlorhydria (absence of hydrochloric acid in the gastric secretions), and current infection by intestinal parasites or bacteria

The clinical symptom appear due to the invading of trophozoites into the wall of colon causing muco-ulation, irregular in shape with submucosal abscesses, and no extra **intestinal balantidiasis**, except in appendix but rarely in liver



***Balantidium* ulcers** may be contaminated by secondary bacterial infection occasionally intestinal perforation, peritonitis, and even death may occur in case of **chronic balantidiasis**.

The patient may suffer from recurrent (persistent) diarrhea alternating with constipation and often have 3-15 motions per day and their stool may contain mucus but usually no blood, no pus cells, and no fever, in acute form, usually with nausea, vomiting, abdominal pain (painful colon) and 3-30 bowel movements per day and the stool contain mucus, blood and on occasion pus cells, dehydration and weight loss are frequent features, masses of trophozoites usually beneath the edge of ulcers in the colon (of irregular shapes).



Cyst of *B. coli* under microscope



Laboratory diagnosis

The best laboratory diagnosis to **balantidiasis** are microscopic examination to patient's stool, which shows cyst stage in formed stool and trophozoite in diarrhea stool. Trophozoite stage motile occur in diarrheic stool and can be easily recognized by their large size, macronucleus, and rapid revolving motility.

Cysts can also be recognized in formed stool of patients by their round shape and presence of large macronucleus and micronucleus.

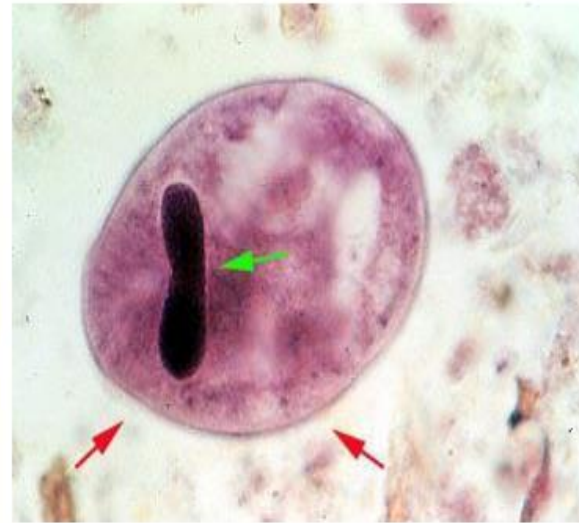


Culture: *B. Coli* may be cultured in vitro in specific media called Locke's egg albumin medium.

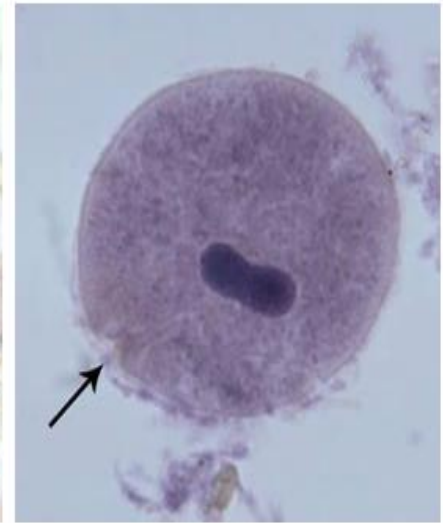
Biopsy: When stool examination is negative, the best laboratory diagnosis by biopsy specimens from muco-ulceration or scraping and stained by periodic acid Schiff (PAS) and microscopic examination shows trophozoites or cyst stage



A



B



C

Microscopic examination of *B. coli* in patient stool

A. trophozoite stage, B&C. cyst stage



Treatment

The best treatment to **balantidiasis** is by **metronidazole (Flagyl)** 500mg/ day four times for ten days with **tetracycline**.

Prophylaxis

- Avoidance of contamination of food and water with humans stool or animal (pork) feces
- Treatment of infected animals and individuals shedding B. coli cyst stage.



Thank you for listening, any questions?